



UNIVERSITY OF
BIRMINGHAM

Superconductivity

Lecture 22 – “Explained II”

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Previously, tunneling

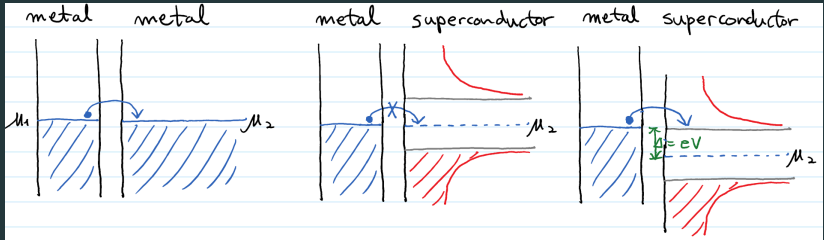
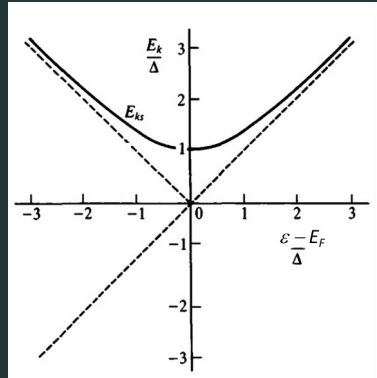
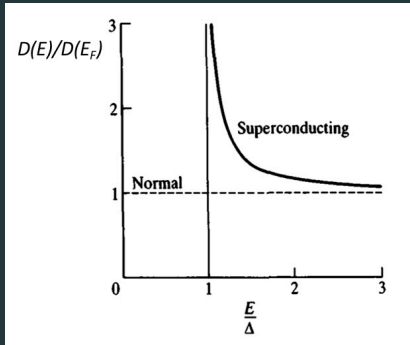


Figure 1: Energy versus density of states for three different contacts of metal-metal, metal-superconductor and superconductor at zero bias, and metal-superconductor at threshold bias.

Previously, density of states



Previously, normal metal to superconductor

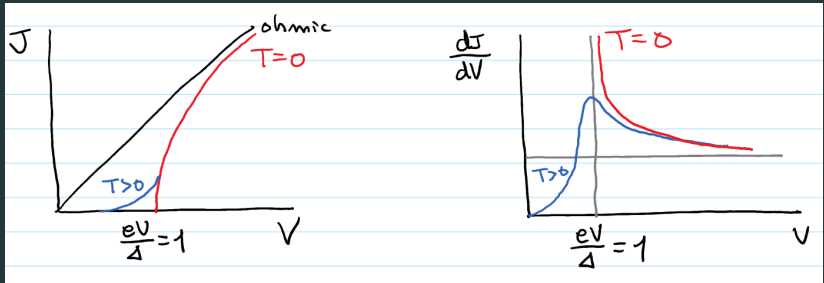


Figure 2: Current-voltage characteristic (left) and differential conductance versus voltage (right).

Conductance ($T = 0$)

$$G_{ns} = \frac{dI}{dV} = G_{nn} \frac{D_s(eV)}{D_n(E_F)} \quad (1)$$

Two superconductors with different gaps ($T = 0$)

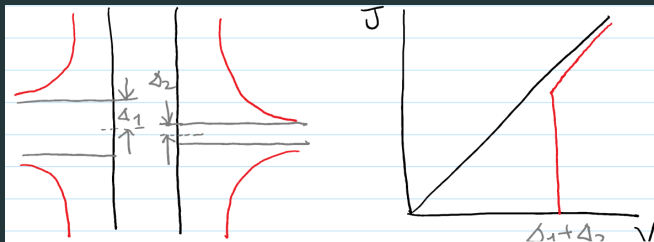


Figure 3: Density of states (left) and I-V characteristic of two different superconductors at zero temperature.

Two superconductors with different gaps ($T \neq 0$)

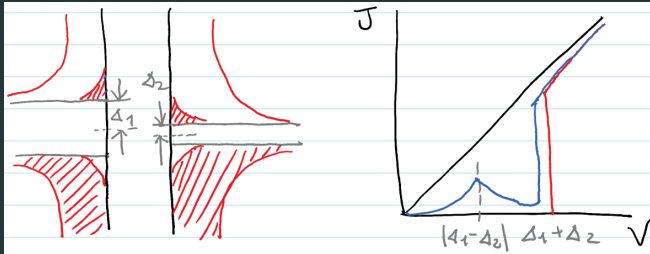


Figure 4: Density of states (left) and I - V characteristic of two different superconductors at finite temperature.

Exercise 1: Tunneling experiment S-I-S

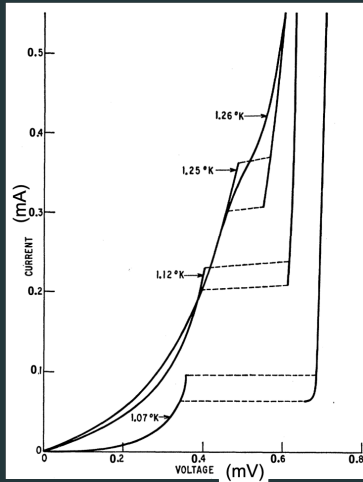


Figure 5: *I-V* tunneling characteristic of a S-I-S junction. (Neglect the hysteric behaviour marked by dashed lines.)

1. Explain why the curve changes its shape as temperature changes in the way as observed.
2. Estimate the values of the most relevant physical quantity of the two superconductors out of which the junction is made.

Exercise 2: Tunneling experiment S-I-S

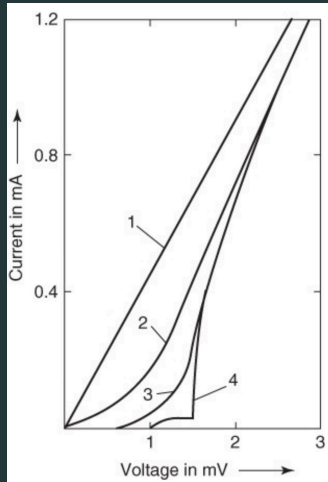


Figure 6: Voltage-current characteristic of an Al-Al₂O₃-Pb tunnel junction measured at different temperatures.

Curve 1 measured at 10 K, curve 2 at 4.2 K, curve 3 at 1.6 K and curve 4 at 1.0 K. Noting that Pb has a higher T_c than Al, use the figure to

1. Estimate Δ at 1.0 K and T_c for both Al and Pb,
2. Briefly explain how it has been estimated.
3. Deduce the relationship between Δ at zero temperature and T_c and briefly justify your answer.

Solution to Exercise 1 (1/2)

The cusp moves to a lower voltage as temperature decreases, and this implies that the gap of one superconductor increases as temperature is lowered while the gap of the other remains rather temperature independent.

Thus in the former superconductor, the measured temperature is not much below T_c . In the latter superconductor, on the other hand, the measured temperature is likely already very low compared to its T_c

(For your interest, the junction was Al-alumina-In, where T_c of bulk Al and In are 1.2 K and 3.4 K, respectively.)

Solution to Exercise 1 (2/2)

At 1.25 K, $\Delta_1 - \Delta_2 = 0.47$ mV and $\Delta_1 + \Delta_2 = 0.5$ mV. So $\Delta_1 = 0.485$ mV and $\Delta_2 = 0.015$ mV.

At 1.12 K, $\Delta_1 - \Delta_2 = 0.40$ mV and $\Delta_1 + \Delta_2 = 0.6$ mV. So $\Delta_1 = 0.5$ mV and $\Delta_2 = 0.1$ mV

At 1.07 K, $\Delta_1 - \Delta_2 = 0.35$ mV and $\Delta_1 + \Delta_2 = 0.68$ mV. So $\Delta_1 = 0.515$ mV and $\Delta_2 = 0.165$ mV

Solution to Exercise 2 (1/2)

At 10 K, V is linear to I , so the temperature is higher than T_c of both superconductors.

At 1.0 K, $|\Delta_1 - \Delta_2| = 1.25$ meV and $|\Delta_1 + \Delta_2| = 1.5$ meV. $\Delta_1 = 1.375$ and $\Delta_2 = 0.125$ meV.

At 4.2 K, V vs I shows deviation from linearity, implying T_c of one of the superconductors is higher than 4.2 K. This allows us to estimate that $T_c = 7(3)$ K for one of the superconductors.

It is only at 1.0 K when the cusp is visible while not clear already at 1.6 K. This allows us to estimate the $T_c = 1.3(3)$ K for the other superconductor.

(For your interest, $T_c = 7.2$ K for Pb and 1.2 K for Al. Compare with our estimation in this exercise, $T_c = 7(3)$ and 1.3(3)K, respectively)

Solution to Exercise 2 (2/2)

To establish a relationship between Δ and T_c , one can only use one of the superconductors which has a significantly higher T_c compared to Δ at 1.0 K that may be approximated as Δ at absolute zero.

Since $k_B T_c = 8.6 \times 10^{-5} \times 7 = 0.6 \text{ meV}$, and $2\Delta = 2.7 \text{ meV}$, one finds that $2\Delta = 4.5 k_B T_c$.

(For your interest, the BCS theory tells us $2\Delta(0) = 3.52 k_B T_c$)